



# secure communication context

“secure” communication

mostly talk about on network

between *principals*  $\approx$  people/servers/programs

but same ideas apply to, e.g., messages on disk  
communicating with yourself

# A to B

running example: A talking with B

maybe sometimes also with C

attacker E — eavesdropper

passive

gets to read all messages over network

attacker M — machine-in-the-middle

active

gets to read and replace and add messages on the network

# privileged network position

intercept radio signal?

control local wifi router?

maybe doesn't just forward messages

compromise network equipment?

send packets with 'wrong' source address  
called "spoofing"

fool DNS servers to 'steal' name?

fool routers to send you other's data?

# possible security properties? (1)

what we'll talk about:

confidentiality — information shared only with those who should have it

authenticity — message genuinely comes from right principal (and not manipulated)

## possible security properties? (2)

important ones we won't talk about...:

repudiation — if A sends message to B, B can't prove to C it came from A

(takes extra effort to get along with authenticity)

forward-secrecy — if A compromised now, E can't use that to decode past conversations with B

anonymity — A can talk to B without B knowing who it is

...

## secrets

if A is talking to B are communicating,  
what stops M (machine-in-the-middle) from pretending to be B?

assumption: B knows some *secret information* that M does not

# secrets

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start: assume A and B have a *shared secret* they both know  
(and attackers do not)

(later: easier to setup assumptions)



# bad ways to use shared secret

A  $\rightarrow$  B: What's the password?

B  $\rightarrow$  A: It's 'Abc\$xyM\$e'.

A  $\rightarrow$  B: That's right! Here's my confidential information.

# bad ways to use shared secret

A  $\rightarrow$  B: What's the password?

B  $\rightarrow$  A: It's 'Abc\$xyM\$e'.

A  $\rightarrow$  B: That's right! Here's my confidential information.

well, this doesn't really help:

- against E (eavesdropper), who can read the password AND confidential info

- against M (machine-in-the-middle), who can also pretend to be A for B

# symmetric encryption

some magic math!

we'll be given two functions by expert:

encrypt:  $E(\text{key}, \text{message}) = \text{ciphertext}$

decrypt:  $D(\text{key}, \text{ciphertext}) = \text{message}$

key = shared secret

ideally small (easy to share) and chosen at random

unsolved problem: how to share it?

# symmetric encryption properties (1)

our functions:

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decrypt:  $D(\text{key}, \text{ciphertext}) = \text{message}$

knowing  $E$  and  $D$ , it should be hard to  
learn anything about the message from the ciphertext without key

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# symmetric encryption procedure

in advance: A and B share encryption key

A computes  $E(\text{key}, \text{'The secret formula is...'}) = ***$

send on network:

A  $\rightarrow$  B: \*\*\*

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# calling things encryption

in this class, (*symmetric*) *encryption* means confidentiality but not authenticity

has malleability problem: attacker can generate a new, valid ciphertext without knowing key.

matches most common thing a library calls encryption

but, sometimes encryption will be “authenticated encryption”  
encryption + message authentication code



# message authentication codes (MACs)

goal: use shared secret *key* to verify message origin

one function:  $MAC(\text{key}, \text{message}) = \text{tag}$

knowing  $MAC$  and the message and the tag, it should be hard to:

- find the value of  $MAC(\text{key}, \text{other message})$  — (“forge” the tag)

- find the key

## contrast: MAC v checksum

message authentication code acts like checksum, but...

checksum can be recomputed without any key

checksum meant to protect against accidents, not malicious attacks

checksum can be faster to compute + shorter

# using MAC without encryption

in advance: choose + share MAC key

A prepares message:

A computes 'Please pay \$100 to M.'

A computes  $MAC(\text{MAC key, 'Please pay \$100 to M.'}) = @@@$

A  $\rightarrow$  B: Please pay \$100 to M. @@@

# using MAC without encryption

in advance: choose + share MAC key

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A computes  $MAC(\text{MAC key, 'Please pay \$100 to M.'}) = @@@$

A  $\rightarrow$  B: Please pay \$100 to M. @@@

B processes message:

B recomputes  $MAC(\text{MAC key, 'Please pay \$100 to M.'})$

*rejects* if it doesn't match @@@

# using MAC with encryption

in advance: choose + share encryption key and MAC key

A prepares message:

A computes  $E(\text{encrypt key, 'The secret formula is...'}) = ***$

A computes  $MAC(\text{MAC key, ***}) = @@@$

A  $\rightarrow$  B: \*\*\* @@@

# using MAC with encryption

in advance: choose + share encryption key and MAC key

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A  $\rightarrow$  B: \*\*\* @@@

B processes message:

B recomputes  $MAC(\text{MAC key, ***})$

*rejects* if it doesn't match @@@

B computes  $D(\text{key, ***}) = \text{'The secret formula is ...'}$

# “authenticated encryption”

often encryption + MAC packaged together

name: authenticated encryption

## exercise

suppose A, B have shared keys  $K_1, K_2$

assume attackers do not have keys

E/D = encrypt/decrypt function

A asks B to pay Sue \$100 by sending message with these parts:

“2023-11-03: pay \$100”

$E(K_1, \text{“2023-11-03 Sue”})$

$MAC(K_2, \text{“2023-11-03: pay $100”})$

1. can eavesdropper learn: (a) who is being paid, (b) how much?
2. can machine-in-middle change: (a) who is being paid, (b) how much?



# shared secrets impractical

problem: shared secrets usually aren't practical

need secure communication before I can do secure communication?

scaling problems

millions of websites  $\times$  billions of browsers = how many keys?

hard to talk to new people

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*need secure communication before I can do secure communication?*

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## *scaling problems*

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# alternative encryption scheme

want to avoid needing to pre-share secret keys

but, ok making some sacrifices:

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*confidentiality in only one direction*

e.g., one set of keys only enables secure communication from  $A \rightarrow B$

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*slower performance*

encryption and decryption can take longer than symmetric encryption

# asymmetric encryption

we'll have two functions:

encrypt:  $PE(\text{public key, message}) = \text{ciphertext}$

decrypt:  $PD(\text{private key, ciphertext}) = \text{message}$

(public key, private key) = "key pair"

# key pairs

‘private key’ = kept secret  
not shared with *anyone*

‘public key’ = safe to give to everyone  
some hard-to-reverse function of private key

concept will appear in some other cryptographic primitives



# asymmetric encryption properties

functions:

encrypt:  $PE(\text{public key, message}) = \text{ciphertext}$

decrypt:  $PD(\text{private key, ciphertext}) = \text{message}$

should have:

knowing  $PE$ ,  $PD$ , the public key, and ciphertext shouldn't make it too easy to find message

knowing  $PE$ ,  $PD$ , the public key, ciphertext, and message shouldn't help in finding private key

# secrecy properties with asymmetric

not going to be able to make things as hard as “try every possible private key”

but going to make it impractical

like with symmetric encryption want to prevent recovery of *any info about message*

also have some other attacks to worry about:

e.g. no info about key should be revealed based on our reactions to decrypting maliciously chosen ciphertexts

# using asymmetric v symmetric

both:

- use secret data to generate key(s)

asymmetric (AKA public-key) encryption

- one “keypair” per recipient

- private key kept by recipient

- public key sent to all potential senders

- encryption is one-way without private key

symmetric encryption

- one key per (recipient + sender)

- secret key kept by recipient + sender

- if you can encrypt, you can decrypt

## using asymmetric encryption

in advance: B generates private key + public key

in advance: B sends public key to A (and maybe others) securely

A computes  $PE(\text{public key, 'The secret formula is...'}) = \text{*****}$

send on network:

A  $\rightarrow$  B: \*\*\*\*\*

B computes  $PD(\text{private key, *****}) = \text{'The secret formula is ...'}$

# digital signatures

symmetric encryption : asymmetric encryption ::  
message authentication codes : digital signatures

# digital signatures

pair of functions:

sign:  $S(\text{private key}, \text{message}) = \text{signature}$

verify:  $V(\text{public key}, \text{signature}, \text{message}) = 1$  (“yes, correct signature”)

$(\text{public key}, \text{private key}) = \text{key pair}$

public key can be shared with everyone

knowing  $S$ ,  $V$ , public key, message, signature

doesn't make it too easy to find another message + signature so that

$V(\text{public key}, \text{other message}, \text{other signature}) = 1$

## using digital signatures

in advance: A generates private key + public key

in advance: A sends public key to B (and maybe others) securely

A computes  $S(\text{private key, 'Please pay ...'}) = \text{*****}$

send on network:

A  $\rightarrow$  B: 'Please pay ...', \*\*\*\*\*

B computes  $V(\text{public key, 'Please pay ...', *****}) = 1$

# replay attacks

A→B: Did you order lunch? [signature 1 by A]

signature 1 by A =  $\text{Sign}(\text{A's private signing key}, \text{"Did you order lunch?"})$

check with  $\text{Verify}(\text{A's public key}, \text{signature 1 by A}, \text{"Did you order lunch?"})$

B→A: Yes. [signature 1 by B]

signature 1 by B =  $\text{Sign}(\text{B's private key}, \text{"Yes."})$

will check with  $\text{Verify}(\text{B's public key}, \text{signature 1 by B}, \text{"Yes."})$

A→B: Vegetarian? [signature 2 by A]

B→A: No, not this time. [signature 2 by B]

...

A→B: There's a guy at the door, says he's here to repair the AC.

Should I let him in? [signature  $N$  by A]



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how can attacker hijack the response to A's inquiry?

*as an attacker, I can copy/paste B's earlier message!*

just keep the same signature, so it can be verified!

Verify(B's public key, "Yes.", signature 2 from B) = 1

# nonces

one solution to replay attacks:

A→B: #1 Did you order lunch? [signature 1 from A]  
signature from A = Sign(A's private key, "#1 Did you order lunch?")

B→A: #1 Yes. [signature 1 from B]

A→B: #2 Vegetarian? [signature 2 from A]

B→A: #2 No, not this time. [signature 2 from B]

...

A→B: #54 There's a guy at the door, says he's here to repair the AC. Should I let him in? [signature  $N$  from A]

(assuming A actually checks the numbers)

## replay attacks (alt)

M→B: #54 Did you order lunch? [signature by M]

B→M: #54 Yes. [signature intended for M by B]

---

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just keep the same signature, so it can be verified!

Verify(B's public key, "#54 Yes.", signature intended for M by B) = 1

## how does A know B's $K_{pub}$ is *actually* B's?

if A asks B for B's public key, M could send M's key instead

could try to verify somehow

e.g., call them on the phone, visit in person

not scalable to do for every website

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*need some way to delegate trust*



## certificate idea

let's say A has B's public key already.

if C wants B's public key and knows A's already:

A can generate a "certificate" certifying B's public key:

"B's public key is XXX" **AND**

Sign(A's private key, "B's public key is XXX")

B sends a copy of this "certificate" to C

if C trusts A, now C has B's public key

if C does not trust A, well, can't trust this either

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# certificate authorities

websites (and others) go to *certificates authorities* (CA) with their public key

certificate authorities sign messages like:  
“The public key for foo.com is XXX.”

signed message called *certificate*

browsers use the certificates to verify website identity

website can forward certificate instead of browser contacting CA directly

# example web certificate (1)

Version: 3 (0x2)

Serial Number: 7b:df:f6:ae:2e:d7:db:74:d3:c5:77:ac:bc:44:bf:1b

Signature Algorithm: sha256WithRSAEncryption

Issuer:

|                        |                          |
|------------------------|--------------------------|
| countryName            | = US                     |
| stateOrProvinceName    | = MI                     |
| localityName           | = Ann Arbor              |
| organizationName       | = Internet2              |
| organizationalUnitName | = InCommon               |
| commonName             | = InCommon RSA Server CA |

Validity

Not Before: Apr 25 00:00:00 2023 GMT

Not After : Apr 24 23:59:59 2024 GMT

Subject:

|                     |                           |
|---------------------|---------------------------|
| countryName         | = US                      |
| stateOrProvinceName | = Virginia                |
| organizationName    | = University of Virginia  |
| commonName          | = canvas.its.virginia.edu |

....

X509v3 extensions:

....

X509v3 Subject Alternative Name: DNS:canvas.its.virginia.edu

# example web certificate (2)

....

Subject Public Key Info:

Public Key Algorithm: rsaEncryption

RSA Public-Key: (2048 bit)

Modulus:

00:a2:fb:5a:fb:2d:d2:a7:75:7e:eb:f4:e4:d4:6c:

94:be:91:a8:6a:21:43:b2:d5:9a:48:b0:64:d9:f7:

f1:88:fa:50:cf:d0:f3:3d:8b:cc:95:f6:46:4b:42:

....

Signature Algorithm: sha256WithRSAEncryption

Signature Value:

24:3a:67:c8:0d:ef:eb:8c:eb:ba:8f:d5:11:d2:1e:ea:44:eb:

fe:af:93:7d:d9:4a:2b:44:a3:7f:47:50:aa:d1:b3:9c:a8:a8:

....

## certificate chains (CA = certificate authority)

canvas.its.virginia.edu has a valid certificate  
that certificate is signed by “InCommon RSA Server CA”

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well, *their* certificate is signed by “USERTrust RSA Certification Authority”

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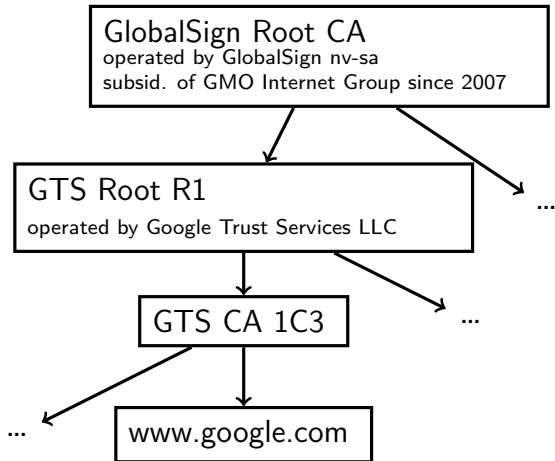
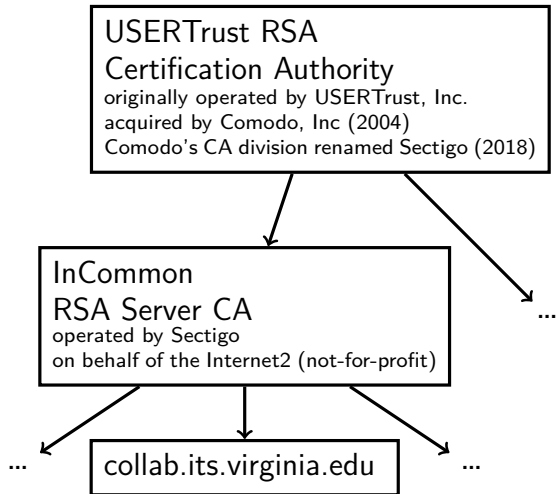
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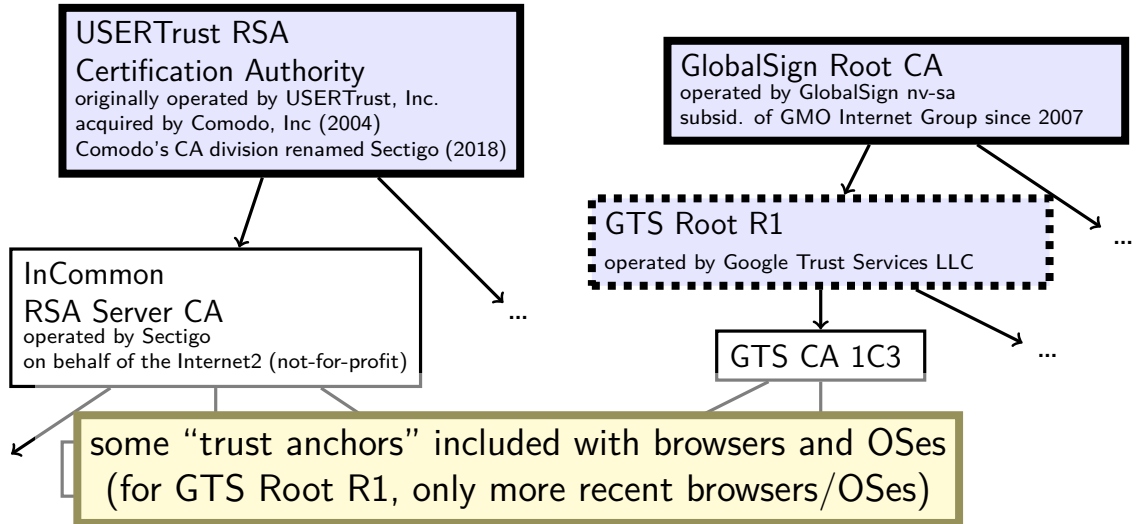
ok, but why do we trust *their* certificate?

*trusted public keys hardcoded in OS/browser*

# certificate hierarchy



# certificate hierarchy



# how many trust anchors?

Mozilla Firefox (as of 27 Feb 2023)

155 trust anchors

operated by 55 distinct entities

Microsoft Windows (as of 27 Feb 2023)

237 trust anchors

operated by 86 distinct entities

Google Chrome (as of 02 Apr 2025)

134 trust anchors

# public-key infrastructure

ecosystem with certificate authorities  
and certificates for everyone

called “public-key infrastructure”

several of these:

- for verifying identity of websites

- for verifying origin of domain name records (kind-of)

- for verifying origin of applications in some OSes/app stores/etc.

- for encrypted email in some organizations

- ...

# additional crypto tools needed for web security

cryptographic hash functions (summarize data)

'secure' random numbers

key agreement



# motivation for hashes: summary for signature

digital signatures typically have size limit

...but we want to sign very large messages

solution: get secure “summary” of message

# cryptographic hash

$$\text{hash}(M) = X$$

given  $X$ : hard to find message other than by guessing

given  $X$ ,  $M$ : hard to find second message so that  $\text{hash}(\text{second message}) = X$

example uses:

- substitute for original message in digital signature
- building message authentication codes

example algorithm: SHA256

# random numbers

need a lot of keys that no one else knows

common task: choose a *random* number

question: what does *random* mean here?

# cryptographically secure random numbers

security properties we might want for random numbers:

attacker cannot guess (part of) number better than chance

knowing prior 'random' numbers shouldn't help predict next 'random' numbers

compromising machine now shouldn't reveal older random numbers

**exercise: how to generate?**

# /dev/urandom

Linux kernel random number generator

collects “entropy” from hard-to-predict events

- e.g. exact timing of I/O interrupts

- e.g. some processor's built-in random number circuit

turned into as many random bytes as you want

# key agreement

problem: A has B's public encryption key  
wants to choose shared secret

some ideas:

- A chooses a key, sends it encrypted to B

- A sends a public key encrypted to B, B chooses a key and sends back

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alternate model (not needed, but usually used by TLS, SSH, ...):

- both sides generate random values

- derive public-key like "key shares" from values

- use math to combine "key shares"

- kinda like A + B both sending each other public encryption keys



# Diffie-Hellman key agreement

A and B want to agree on shared secret

A chooses random value  $Y$

A sends public value derived from  $Y$  (“key share”)

B chooses random value  $Z$

B sends public value derived from  $Z$  (“key share”)

A combines  $Y$  with public value from B to get number

B combines  $Z$  with public value from A to get number  
and b/c of math chosen, both get same number

# Diffie-Hellman key agreement (details, if needed)

math requirement:

some  $f$ , so  $f(f(X, Y), Z) = f(f(X, Z), Y)$   
(that's hard to invert, etc.)

choose  $X$  in advance and:

|                            |                            |
|----------------------------|----------------------------|
| A randomly chooses $Y$     | B randomly chooses $Z$     |
| A sends $f(X, Y)$ to B     | B sends $f(X, Z)$ to A     |
| A computes $f(f(X, Z), Y)$ | B computes $f(f(X, Y), Z)$ |

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| A sends $f(X, Y)$ to B     | B sends $f(X, Z)$ to A     |
| A computes $f(f(X, Z), Y)$ | B computes $f(f(X, Y), Z)$ |

example  $f(a, b) = a^b \pmod{p}$

# why not just asymmetric encryption?

given public-key encryption + digital signatures...

why bother with the symmetric stuff?

symmetric stuff much faster

symmetric stuff much better at supporting larger messages

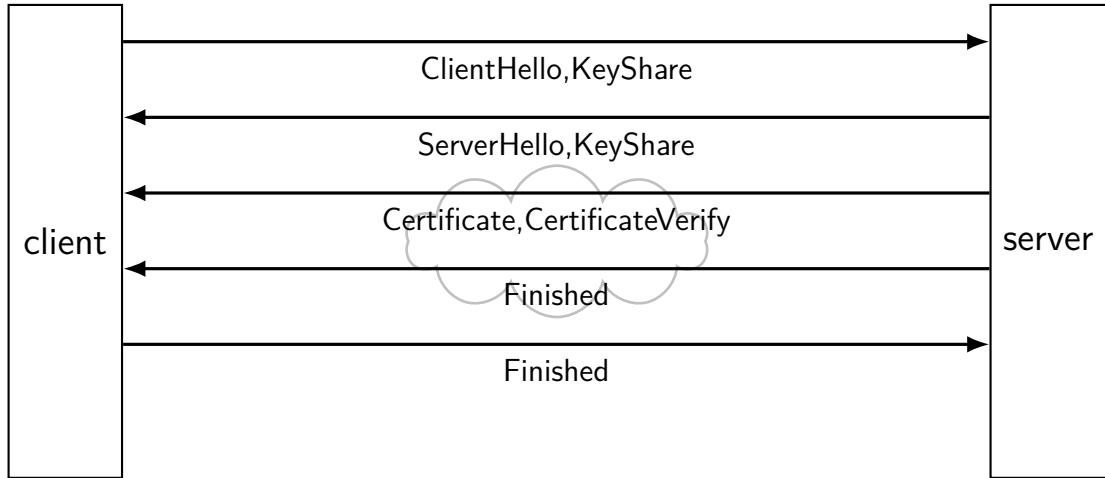
# TLS: Transport Layer Security

the secure communication protocol for the web

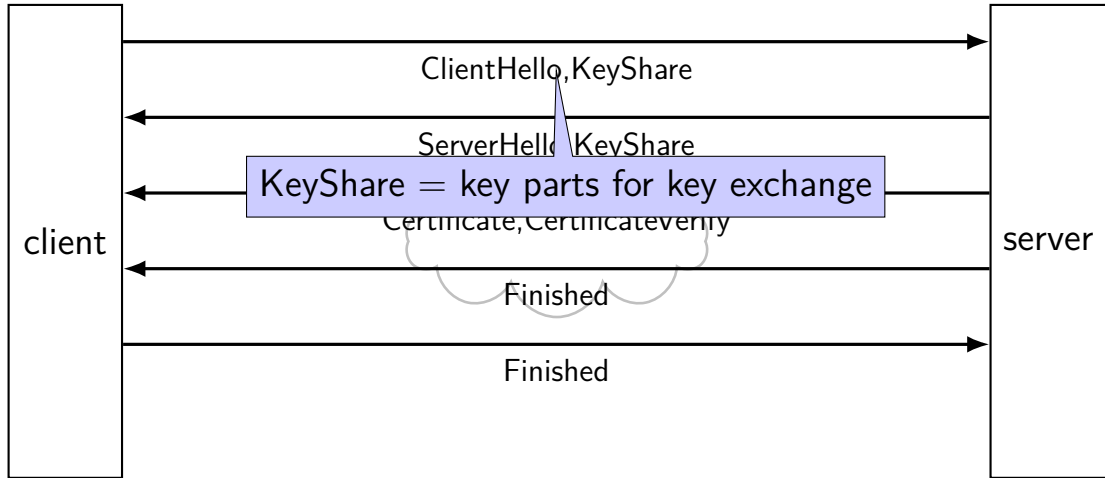
what makes accessing websites secure

the "s" in *https*

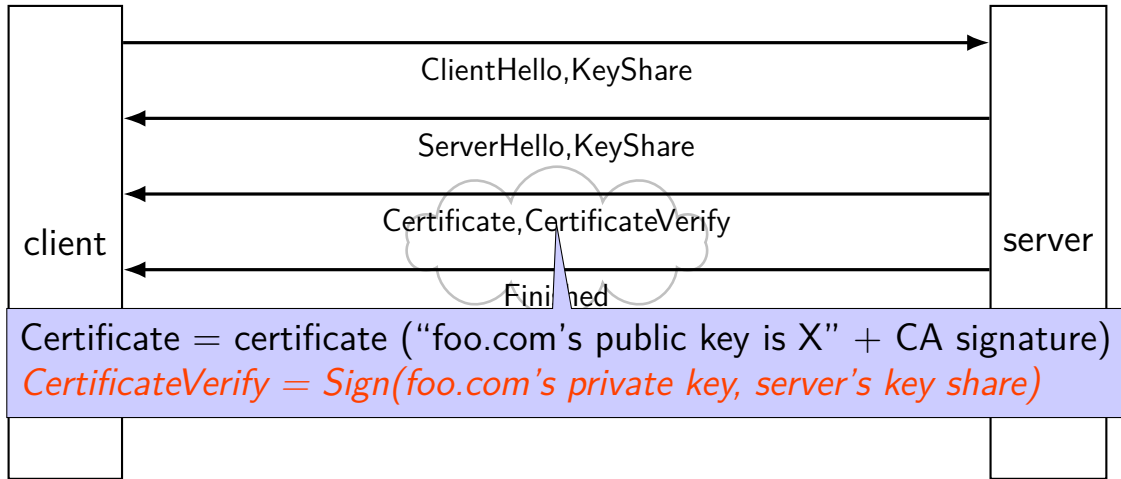
# typical TLS handshake



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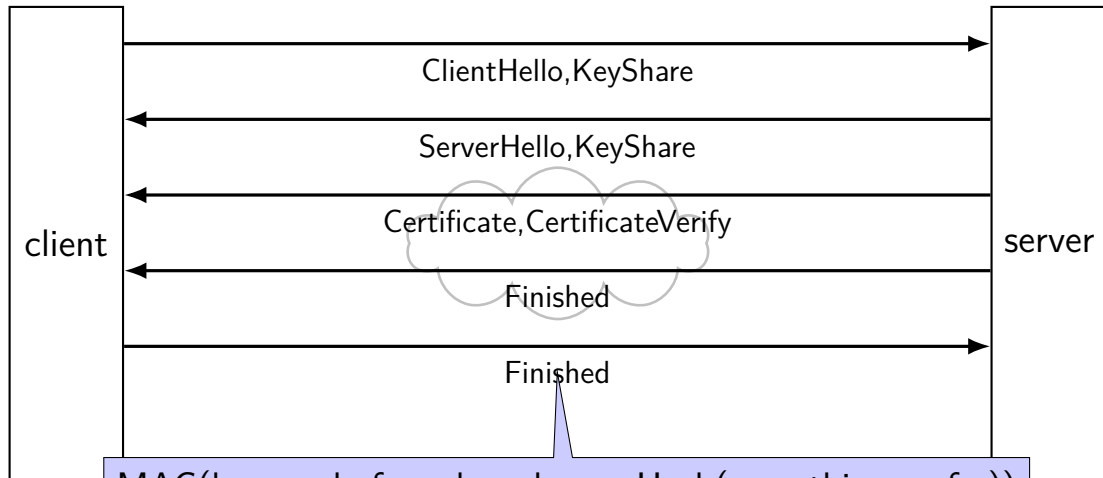
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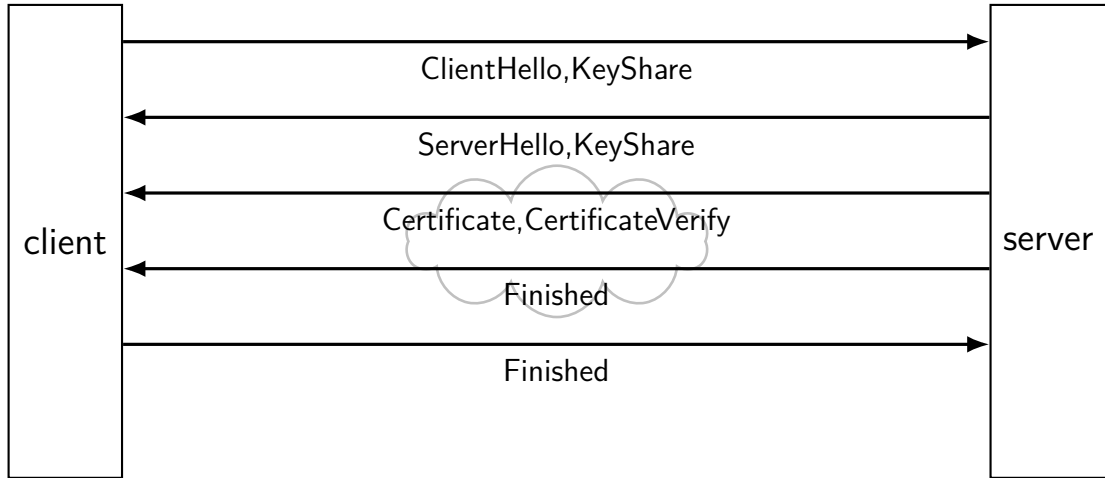


# typical TLS handshake



MAC(key made from key shares, Hash(everything so far))  
(purpose: tie new key with rest of handshake)

# typical TLS handshake



# TLS: after handshake

use key shares results to get **several** symmetric keys

take  $\text{hash}(\text{something} + \text{shared secret})$  to derive each key

separate keys for each direction (server  $\rightarrow$  client and vice-versa)

often separate keys for encryption and MAC

later messages use encryption + MAC + nonces

# things modern TLS usually does

(not all these properties provided by all TLS versions and modes)

confidentiality/authenticity

- server = one ID'd by certificate

- client = same throughout whole connection

forward secrecy

- can't decrypt old conversations (data for KeyShares is temporary)

fast

- most communication done with more efficient symmetric ciphers

- 1 set of messages back and forth to setup connection

# network security summary (1)

communicating securely with math

- secret value (shared key, public key) that attacker can't have

- symmetric: shared keys used for (de)encryption + auth/verify; fast

- asymmetric: public key used by any for encrypt + verify; slower

- asymmetric: private key used by holder for decrypt + sign; slower

protocol attacks — repurposing encrypt/signed/etc. messages

certificates — verifiable forwarded public keys

key agreement — for generated shared-secret “in public”

- publish key shares from private data

- combine private data with key share for shared secret

## network security summary (2)

TLS: combine all cryptography stuff to make “secure channel”



# backup slides

## tools, but...

have building blocks, but less than straightforward to use

lots of issues from using building blocks poorly

start of art solution: formal proof systems

- mathematical proof that attacker doing X implies encryption/MAC/etc.

- broken

- ideally a somewhat machine-checkable proof

(we aren't going to be that formal...)

# other attacks without breaking math

# TLS state machine attack

from <https://mitls.org/pages/attacks/SMACK>

protocol:

- step 1: verify server identity
- step 2: receive messages from server

attack:

- if server sends “here’s your next message”,  
instead of “here’s my identity”  
then broken client ignores verifying server’s identity

# Matrix vulnerabilities

one example from <https://nebuchadnezzar-megolm.github.io/static/paper.pdf>

system for confidential multi-user chat

protocol + goals:

- each device (my phone, my desktop) has public key
- to talk to me, you verify one of my public keys
- to add devices, my client can forward my other devices' public keys

bug:

- when receiving new keys, clients did not check who they were forwarded from correctly

## exercise

exercise: how should website certificates verify identity?

# how do certificate authorities verify

for web sites, set by CA/Browser Forum

organization of:

- everyone who ships code with list of valid certificate authorities

  - Apple, Google, Microsoft, Mozilla, Opera, Cisco, Qihoo 360, Brave, ...

- certificate authorities

decide on rules (“baseline requirements”) for what CAs do

# BR domain name identity validation

options involve CA choosing random value and:

sending it to domain contact (with domain registrar) and receive response with it, or

observing it placed in DNS or website or sent from server in other specific way

exercise: problems this doesn't deal with?



## some other things public CAs do

- keep their private keys in tamper-resistant hardware

- maintain publicly-accessible database of *revoked* certificates
  - some browsers check these, sometimes

- certificate transparency

  - public logs of every certificate issued
  - some browsers reject non-logged certificates
  - so you can tell if bad certificate exists for your website

- 'CAA' records in the domain name system

  - can indicate which CAs are allowed to issue certificates in DNS  
(but CAs apparently not required to use DNSSEC (certificate infrastructure for signing domain name records) when looking this up)

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# password hashing

cryptographic hash functions need (basically) guessing to 'reverse'

idea: store cryptographic hash of password instead of password

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fix: special slow/resource-intensive cryptograph hash functions

Argon2i

scrypt

PBKDF2

# key agreement and asym. encryption

can construct public-key encryption from key agreement

private key: generated random value  $Y$

public key: key share generated from that  $Y$



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- combine with public key to get shared secret

- use symmetric encryption + MAC using shared secret as keys

- output: (key share generated from  $Z$ ) (sym. encrypted data) (mac tag)

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combine with public key to get shared secret  
use symmetric encryption + MAC using shared secret as keys  
output: (key share generated from  $Z$ ) (sym. encrypted data) (mac tag)

$PD(\text{private key, message}) =$   
extract (key share generated from  $Z$ )  
combine with private key to get shared secret, ...

# denial of service (1)

so far: worried about network attacker disrupting confidentiality/authenticity

what if we're just worried about just breaking things

well, if they control network, nothing we can do...

but often worried about less

## denial of service (2)

if you just want to inconvenience...

attacker just sends lots of stuff to my server

my server becomes overloaded?

my network becomes overloaded?

but: doesn't this require a lot of work for attacker?

exercise: why is this often not a big obstacle

# denial of service: asymmetry

work for attacker  $>$  work for defender

how much computation per message?

- complex search query?

- something that needs tons of memory?

- something that needs to read tons from disk?

how much sent back per message?

resources for attacker  $>$  resources of defender

how many machines can attacker use?

# denial of service: reflection/amplification

instead of sending messages directly...attacker can send messages  
“from” you to third-party

third-party sends back replies that overwhelm network

example: short DNS query with lots of things in response

“amplification” =

third-party inadvertently turns small attack into big one

# firewalls

don't want to expose network service to everyone?

solutions:

- service picky about who it accepts connections from
- filters in OS on machine with services
- filters on router

later two called “firewalls”

# firewall rules examples?

ALLOW tcp port 443 (https) FROM everyone

ALLOW tcp port 22 (ssh) FROM *my desktop's IP address*

BLOCK tcp port 22 (ssh) FROM everyone else

ALLOW from address X to address Y

...



# backup slides

# cryptographic hash uses

find shorter 'summary' to substitute for data  
what hashtables use them for, but...  
we care that adversaries can't cause collisions!

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deal with message limits in signatures/etc.

password hashing — but be careful! [next slide]

constructing message authentication codes  
hash message + secret info (+ some other details)